

SCHOOLCRAFT COUNTY AP, MI, USA

WMO: 725408

Lat: 45.975N

Lon: 86.172W

Elev: 209

StdP: 98.84

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04881

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-18.4	-26.2	0.4	-19.4	-23.2	0.5	-17.4	11.0	-5.1	9.5	-5.4	3.3	290	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	26.9	20.1	25.0	19.7	23.7	19.1	22.3	24.7	21.3	23.6	20.2	22.6	3.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.4	23.9	20.5	15.6	23.0	19.1	14.2	21.7	66.4	24.7	63.0	23.4	59.0	22.5	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.3	7.5	-24.7	29.9	3.6	2.2	-27.3	31.5	-29.4	32.7	-31.4	33.9	-34.0	35.5		
(5)			-25.0	23.8	3.5	1.3	-27.5	24.7	-29.6	25.4	-31.5	26.1	-34.1	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-7.3	-7.3	-2.6	3.5	9.4	14.7	18.0	17.9	14.2	7.7	1.9	-4.2		
	DBStd	10.12	5.80	5.62	5.59	4.06	3.67	3.20	3.09	2.88	3.91	4.16	4.64	5.28		
	HDD10.0	2461	538	484	392	196	55	3	0	0	9	98	245	440		
	HDD18.3	4739	796	717	650	444	277	114	44	43	131	331	494	698		
	CDD10.0	841	0	0	1	2	37	145	249	244	136	27	1	0		
	CDD18.3	78	0	0	0	0	1	6	34	29	8	0	0	0		
	CDH23.3	363	0	0	0	0	4	34	191	121	13	0	0	0		
	CDH26.7	43	0	0	0	0	1	4	29	10	0	0	0	0		
(14) Wind	WSAvg	3.4	3.7	3.7	3.7	3.9	3.6	3.0	2.9	2.9	3.1	3.5	3.6	3.6		
(15) Precipitation	PrecAvg	780	44	31	50	64	78	78	77	75	88	81	63	48		
	PrecMax	1003	144	81	124	123	142	172	171	113	207	148	122	100		
	PrecMin	620	17	4	8	20	41	25	16	22	29	31	24	6		
	PrecStd	96	27	18	32	26	26	35	41	21	36	33	27	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	6.2	12.8	17.2	22.9	27.2	29.1	27.7	25.1	19.8	13.8	8.7		
		MCWB	3.2	5.4	8.6	9.2	14.7	18.8	20.6	20.8	19.6	16.8	10.5	8.1		
	2%	DB	2.9	3.0	8.9	13.9	20.1	23.8	27.3	26.2	22.9	17.8	11.7	6.1		
		MCWB	1.8	2.0	6.0	7.4	14.1	17.5	20.4	20.4	19.3	15.3	9.9	4.9		
	5%	DB	2.2	2.1	6.3	12.2	17.8	22.3	25.9	24.5	22.1	16.8	9.9	4.1		
		MCWB	1.2	1.3	3.7	6.6	12.6	16.8	20.0	19.9	18.7	14.3	8.4	2.9		
	10%	DB	1.0	0.9	4.0	10.0	16.1	20.9	23.8	23.1	20.9	14.9	8.6	2.8		
		MCWB	0.2	-0.1	2.1	5.9	11.5	16.4	19.3	19.1	18.1	12.7	7.4	1.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	5.6	10.4	11.2	17.8	21.0	23.5	23.6	21.5	17.8	11.8	8.0		
		MCDB	4.5	6.9	12.2	14.1	19.8	23.8	26.6	26.0	23.3	18.6	12.9	8.2		
	2%	WB	2.4	2.6	6.6	9.1	15.5	19.4	22.4	22.4	20.3	16.1	10.2	5.1		
		MCDB	3.1	3.0	8.5	12.2	18.3	22.2	25.2	24.6	22.0	17.5	11.3	6.1		
	5%	WB	1.1	1.1	4.2	7.9	13.8	18.1	21.3	21.4	19.4	14.6	8.5	3.5		
		MCDB	1.8	1.7	6.2	10.4	16.6	20.5	24.0	23.5	21.1	16.4	9.3	4.3		
	10%	WB	0.0	-0.3	2.7	6.6	12.3	17.1	20.2	20.3	18.4	13.0	7.5	2.2		
		MCDB	0.7	0.5	4.0	9.4	14.9	19.2	22.9	22.5	19.9	14.5	8.2	2.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	8.1	8.3	8.8	9.9	10.0	10.3	10.0	9.4	8.2	6.7	6.7		
		MCDBR	6.1	6.7	8.8	12.3	12.7	12.0	11.8	10.8	9.5	9.5	8.1	6.7		
	5% WB	MCWBR	5.7	5.8	6.4	7.4	7.6	7.1	6.8	6.9	6.6	7.3	6.8	6.4		
		MCWBR	5.9	6.5	8.1	10.2	9.9	9.6	9.8	9.2	7.9	8.1	7.5	6.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.268	0.281	0.309	0.353	0.381	0.395	0.391	0.385	0.366	0.333	0.302	0.270		
		taud	2.403	2.347	2.383	2.367	2.353	2.345	2.371	2.389	2.440	2.513	2.518	2.452		
	Edn at Noon	Edn at Noon	846	902	922	905	886	872	870	863	850	834	802	806		
		Edh at Noon	70	91	102	114	120	122	117	111	97	78	64	61		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.18	2.09	3.44	4.54	5.57	6.01	6.13	5.19	3.90	2.32	1.26	0.90		
	RadStd	RadStd	0.11	0.25	0.34	0.46	0.43	0.45	0.28	0.34	0.29	0.26	0.15	0.08		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page